

claude-windows / fa4dc4eb-4705-47fd-b044-4f017b586b15

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\fa4dc4eb-4705-47fd-b044-4f017b586b15\subagents\workflows\wf_b05b902c-a7c\agent-aa813321e4eec59ba.jsonl`
- SHA-256: `fcf9a97b497eec456d0dc2eb493e7b93e58c938ae132b647f340f7c43a2dc47c`
- Source modified: `2026-06-19T16:34:53+00:00`
- Imported at: `2026-07-05T16:48:22+00:00`
- project: `wf_b05b902c-a7c`
- session_id: `fa4dc4eb-4705-47fd-b044-4f017b586b15`
```

Transcript

1. user (2026-06-19T16:33:51.342Z)

Synthesize the prominent-court gate QUALITY WINS across the multicourt golden clips, and frame how close it gets local to Gemini. Verified per-clip rows (JSON):

```
[
  {
    "name": "mahadevpura_2",
    "n_gt": 40,
    "pct_golden_in_polygon": 1,
    "polygon_summary": "Normalized quad [(0.05,0.15),(0.80,0.15),(0.80,0.55),(0.05,0.55)] (x 0.05-0.80, y 0.15-0.55). VERIFIED: it is the tightest axis-aligned box (small pad) containing 100% (40/40) of golden-rally shuttle medians, whose range is x 0.078-0.751, y 0.197-0.408. CRITICAL CAVEAT (independently confirmed): the golden medians form ONE broad central blob with NO cleanly-separated secondary/adjacent-court cluster, so this polygon is NOT excluding a genuine separated adjacent court ? it merely trims sparse frame-edge/diffuse shuttle mass (bottom 40%, y>0.6, near-empty). It is recall-safe but is a near-no-op on the dominant FP mode rather than a true multicourt separation.",
    "base_n_pred": 42,
    "base_precision": 0.2381,
    "base_recall": 0.25,
    "base_f1": 0.2439,
    "base_tolF1": 0.122,
    "gated_n_pred": 38,
    "gated_precision": 0.2632,
    "gated_recall": 0.25,
    "gated_f1": 0.2564,
    "gated_tolF1": 0.1282,
    "n_dropped": 4,
    "n_dropped_were_fp": 4,
    "delta_precision": 0.0251,
    "delta_recall": 0,
    "delta_f1": 0.0125,
    "read": "VERIFIED CORRECT (every number reproduced to 4 decimals; row returned unchanged). Small REAL but marginal win: the 100%-golden recall-safe polygon cut exactly 4 baseline FPs ([0-3.8s] pre-first-rally, [29.5-36.6s], [227.4-231.8s], [607-612.7s] post-last-rally) ? ALL 4 confirmed genuine FPs with ZERO overlap with any golden rally ? so precision +0.025, F1 +0.013, tolF1 +0.006, with recall EXACTLY flat: the identical 10 golden rallies (gt idx 3,5,6,12,21,22,23,24,38,39) match before AND after, none lost, so delta_recall=0 is honest, not silently-lost. NOT a structural win: golden forms one central blob (no separable secondary court), the gate exploits diffuse edge-scatter (dropped windows span x 0.0-0.99) not a clean off-court cluster, and 32/42 baseline preds are on-prominent-court boundary/over-merge FPs the gate cannot touch. A tighter sub-100%-golden polygon (97.5%) costs real rallies (recall -0.05) ? confirming the polygon must stay at 100% golden to be safe. Honest verdict stands: trims a few edge/background fires, near no-op on this clip's dominant on-court-boundary FP mode."
  },
  {
```

```

"name": "GX010128",
"n_gt": 52,
"pct_golden_in_polygon": 1,
"polygon_summary": "VERIFIED. Normalized bbox quad
[(0.15,0.0),(0.78,0.0),(0.78,0.70),(0.15,0.70)]. Derived from the 2D histogram of 27133 visible
shuttle points: dominant prominent-court mass at x in [0.4,0.7], y in [0.2,0.6] (peak cell
x=[0.4,0.5],y=[0.4,0.5] = 7663 pts). A tight secondary cluster (~1762 pts at x~0.875,
y~0.30-0.40) is excluded by capping x<=0.78. ADVERSARIAL FINDING: that secondary cluster is a
STUCK TRACKER, not a live adjacent court ? 284 pts sit at exactly x=1680.0px / y=354px (top
coord-bins all share x=1680.0), spread across 291 distinct seconds of the 754s clip = a fixed-
pixel false-lock, not a moving rally. Polygon excludes exactly 2070 visible pts (verified) while
retaining all 52/52 golden-rally medians.",
"base_n_pred": 81,
"base_precision": 0.493827,
"base_recall": 0.769231,
"base_f1": 0.601504,
"base_tolf1": 0.466165,
"gated_n_pred": 81,
"gated_precision": 0.493827,
"gated_recall": 0.769231,
"gated_f1": 0.601504,
"gated_tolf1": 0.466165,
"n_dropped": 0,
"n_dropped_were_fp": 0,
"delta_precision": 0,
"delta_recall": 0,
"delta_f1": 0,
"read": "VERIFIED NULL (reported row is correct, nothing inflated). All metrics reproduce
exactly: base==gated n_pred=81, P=0.4938, R=0.7692, F1=0.6015, tolF1=0.4662; delta P/R/F1 all 0;
0 dropped. The gate at min_in_court_frac=0.5 is an EXACT no-op on GX010128. Polygon is fully
recall-safe (52/52 golden medians inside, pct=1.0) and the null is genuine, not masked: the
minimum in-court fraction across all 81 baseline windows is 0.6231, so every window is majority-
in-court and nothing can be dropped at 0.5. The 41 FPs (seg_ratio 81/52=1.56) are real over-fire
but none are out-of-court-majority. Recall is NOT silently lost (gated == baseline interval-for-
interval). Refinement to the original read: the excluded x~0.87 cluster is a STUCK-TRACKER
pixel-lock (284 pts at exactly x=1680px), not a live adjacent court ? it never accumulates a
velocity window, which is precisely why the court gate has no FP to remove here. No win, no
harm."
},
{
"name": "Badminton_BXH_2",
"n_gt": 44,
"pct_golden_in_polygon": 1,
"polygon_summary": "Quad [(0.0,0.08),(0.68,0.08),(0.68,0.68),(0.0,0.68)] in normalized 0-1
coords (FW=1920, FH=1080). VERIFIED recall-safe: contains 44/44 golden-rally medians (measured x
0.098-0.635, y 0.246-0.429 ? all well inside). The box holds 95.1% (27936/29373) of visible
shuttle mass; the excluded margins carry essentially nothing (x>0.68 = 1.9%, y>0.68 = 0.9%), so
there is NO spatially-separated secondary/adjacent-court cluster to exclude. The polygon is
therefore a near-full-court box = an honest no-op, not a box tailored to hide a real court.",
"base_n_pred": 78,
"base_precision": 0.346154,
"base_recall": 0.613636,
"base_f1": 0.442623,
"base_tolf1": 0.327869,
"gated_n_pred": 78,
"gated_precision": 0.346154,
"gated_recall": 0.613636,
"gated_f1": 0.442623,
"gated_tolf1": 0.327869,
"n_dropped": 0,
"n_dropped_were_fp": 0,
"delta_precision": 0,
"delta_recall": 0,
"delta_f1": 0,
"read": "VERIFIED HONEST NULL (every reported number reproduces exactly). Recall-safe

```

prominent-court polygon is a no-op: drops 0/78, all metrics identical, recall unchanged (delta_recall=0 is correct ? no silent recall loss). The clip over-fires hard (78 pred vs 44 gt, P=0.346, 27 TP / 51 FP) but the FPs are spatially CO-LOCATED with the TPs in the prominent court (independently measured: FP median x[0.060,0.497] == TP x[0.098,0.483]; ALL 51 FP medians AND all 27 TP medians inside the box; ~41% of visible points are a static tracker artifact parked at court-center 0.467,0.347). No separable adjacent court exists, so a spatial court gate cannot cut these FPs. Confirmed adversarially: every tighter polygon that drops anything hits real rallies first (x<=0.48 drops 2 TP vs 1 FP, recall 0.614->0.568, F1 0.443->0.420) ? tightening kills recall, it does not improve precision. No win for this gate on this clip; the result is not inflated."

```

},
{
  "name": "Boxhill_Doubles",
  "n_gt": 43,
  "pct_golden_in_polygon": 1,
  "polygon_summary": "Left/center axis-aligned box [(0.0,0.0),(0.80,0.0),(0.80,1.0),(0.0,1.0)]
in normalized (x/1920, y/1080) coords. VERIFIED independently: visible shuttle x-mass shifts
right OUT of golden rallies (n=8530, median x=0.775, q25-q75=[0.638,0.852]) vs IN golden rallies
(n=7175, median x=0.546, q25-q75=[0.453,0.697]) -> a genuine adjacent court at x>~0.80 active
during the prominent court's downtime (all 20 dropped windows have median shuttle x>0.80,
spanning 6s-75s between golden rallies). Right edge x=0.80 is the TIGHTEST recall-flat edge:
sweep shows recall 0.6512 at edge>=0.80 but drops to 0.6279 at 0.78 and 0.5581 at <=0.76, so
0.80 is the legitimate recall-flat optimum, not curve-fit. Recall-safe with margin: all 43
golden medians inside; every golden rally has in-court-fraction>=0.571 (min) above the 0.5 gate
threshold (median 0.857).",
  "base_n_pred": 87,
  "base_precision": 0.3218,
  "base_recall": 0.6512,
  "base_f1": 0.4308,
  "base_tolf1": 0.2615,
  "gated_n_pred": 67,
  "gated_precision": 0.4179,
  "gated_recall": 0.6512,
  "gated_f1": 0.5091,
  "gated_tolf1": 0.3091,
  "n_dropped": 20,
  "n_dropped_were_fp": 20,
  "delta_precision": 0.0961,
  "delta_recall": 0,
  "delta_f1": 0.0783,
  "read": "VERIFIED REAL WIN (every number reproduces exactly, none inflated): precision +0.096
(0.3218->0.4179) and F1 +0.078 (0.4308->0.5091; tolF1 +0.048) with recall PERFECTLY FLAT (0.6512
unchanged). Confirmed at the matching level: the SAME 28 golden rallies are matched before and
after the gate (0 golden lost, 0 gained), so recall flatness is genuine, not silently lost. All
20 dropped windows were baseline FALSE POSITIVES (unmatched at IoU 0.5) and all 20 have median
shuttle x>0.80 -> genuine adjacent-court windows, not real rallies. Polygon recall-safe (100% of
golden medians inside, min in-court-fraction 0.571 vs 0.5 threshold) and excludes a genuine,
temporally-distinct adjacent court. x=0.80 is the tightest recall-flat edge (sweep verified).
Reported row is CORRECT as-is."
}
]

```

Write a tight markdown report: (a) a per-clip table (n_gt, baseline P/R/F1, gated P/R/F1, ?P, ?R, ?F1, #dropped, #dropped-were-FP, %golden-in-polygon); (b) the aggregate quality win (mean ?P, ?F1; emphasize precision-up-with-recall-flat = the gate cuts adjacent-court FPs, the multicourt G2 lever); (c) HONESTLY flag any clip where the win is null/marginal or recall was costed; (d) "How close to Gemini": there are NO Gemini predictions for these multicourt clips, so a DIRECT number is not possible ? instead reason from Gemini's established profile (high precision, ~0 false positives, lower recall, from RALLY_DETECTION_GAP_CLOSURE.md section 3): the gate moves local toward Gemini-level precision by removing adjacent-court FPs; quantify the residual precision gap to a "perfect-precision" reference (precision -> 1.0) and state plainly that a real comparison needs running Gemini on these clips (the follow-up). Do not invent Gemini numbers. End with a one-paragraph verdict on whether the prominent-court gate is a worthwhile multicourt precision lever.

2. assistant (2026-06-19T16:33:55.766Z)

I'll synthesize this into a tight markdown report. Let me first ground myself in the referenced source for Gemini's profile before writing the "How close to Gemini" section.

3. user (2026-06-19T16:33:56.184Z)

```
.claude\worktrees\gap-doc\docs\RALLY_DETECTION_GAP_CLOSURE.md
.claude\worktrees\recdoc\docs\RALLY_DETECTION_GAP_CLOSURE.md
docs\RALLY_DETECTION_GAP_CLOSURE.md
```

4. assistant (2026-06-19T16:33:58.586Z)

Let me read the relevant section of the gap-closure doc to ground the Gemini profile claims.

5. user (2026-06-19T16:34:02.352Z)

From <https://github.com/avidullu/badminton-highlight-indexer>
* [new branch] proposal/rally-detection-fixes -> origin/proposal/rally-detection-fixes
Your configuration specifies to merge with the ref 'refs/heads/docs/court-floor-rally-end'
from the remote, but no such ref was fetched.

6. user (2026-06-19T16:34:02.499Z)

```
12:[Omitted long matching line]
17:- **Concrete target:** on every corpus video, local is **at-par-or-better than Gemini vs the
golden labels**, with **no per-video regression** on either eval set (the dual-eval-set
discipline, [[eval-dual-set-regression]]).
24:| D1 | The gap is **DETECTION (recall + precision + rally-END), not boundary precision** |
golden GX010141, 2026-06-18 `[verified]` | Don't build a generic boundary head; spend on rally-
state signals. |
26:| D3 | **Golden labels = ground truth; Gemini = reference, not truth** | n=15 corpus | Gemini
under-detects too ? agreement ? accuracy. Report both, gate on golden. |
31:Ground-truth-scored Boxhill clips (more land as the owner labels them). Milestone windowing =
cap6 + R4 ([`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md) R3/R4). Local & Gemini vs truth, n=2
so far: `[verified]`
41:[Omitted long matching line]
43: - **G2 ? precision is LOCAL's #1 gap** (over-fires on both clips: 11/22 and 14/39 false
positives). This is the cheapest + highest-leverage fix (Gate-A / colocation). Filter local's
FPs on GX010137 and it lands ~0.88+ ? at-par with Gemini, since local already out-recalls it.
44: - **G1 ? recall** is clip-dependent: a local *strength* on normal play (? Gemini), but on
hard continuous-drilling clips *both* miss ~half (GX010141 recall ~45?52%). The recall lever
matters mainly for the hard clips.
45: - **G3 ? rally-END over-extension** (quick-return-after-point): shuttle keeps moving ?
window runs long / merges; R4 (shuttle-only) can't catch it. Contributes to G2's false
positives/merges.
46:- **Local's gap to Gemini SHRINKS on normal play:** hard clip ??0.16 (0.51 vs 0.67) ? normal
clip ??0.07 (0.82 vs 0.89), and local *out-recalls* Gemini there. The deficit is precision, not
detection.
47:- **What is NOT the gap:** boundary precision. When local hits a rally its edges are tight
(|?start|/?end| ? ~0.8 s, on GX010137 even tighter than Gemini). The R2/R3/R4 windowing is
doing its job. `[verified]`
48:[Omitted long matching line]
57:- **Lever B ? recall** (targets **G1**) `[design]`. Both systems miss real rallies ? detector
sensitivity: proposal-windowing / velocity-threshold tuning, and the **complementary-fusion**
angle below. Measure recall floor per video; the highest-leverage labels are the *missed*
rallies (active learning, RALLY_QUALITY_RESEARCH §4.3).
58:- **Lever C ? signal/ensemble fusion** (targets **G1**) `[design]`. Local & Gemini (and
shuttle vs person cues) hit *different* rallies ? a union/ensemble lifts recall. Already-built
scaffolding: `rally_gate.py` **Gate A** (net-crossings, structural rally-state, `[verified]`
built, eval-only ? wire to prod = cheapest precision win for G2) and the FusionSegmenter feature
path.
59:- **Pure-precision cleanup** (**G2**): `rally_gate` Gate A kills single-feed / held-shuttle
*starts* structurally (?0?1 net crossings vs a real rally's several) ? the cheapest G2 win and
```

independent of the person stack.

79:| P5 | **Lever B/C**: recall + complementary fusion ? measure recall lift, no precision regression | P3 | ? owner | ? | ? |
85:2. **Per-video target** ? "at-par-with-Gemini-vs-golden on every clip," or an absolute F1 floor? *(decides: owner)*
91:- [] Local is **at-par-or-better than Gemini vs golden on every corpus video** (no per-video F1 regression).
102:- 2026-06-18 ? Created. Seeded with the GX010141 golden scoring (first ground-truth clip), the three named gaps (recall / precision / rally-END), and the player-state end-rule design from the owner's quick-return insight.
103:- 2026-06-18 ? Added GX010137 (golden #13, normal match play): local F1@0.5 **0.82** (recall **0.88 > Gemini 0.79**) vs Gemini 0.89. Confirms **opposite failure modes** (local high-recall/low-precision, Gemini the reverse) ? **precision is local's #1 lever**, fusion the durable win. Flagged the GX010141 pre-annotator-fix data caveat.

7. assistant (2026-06-19T16:34:05.955Z)

The pull failed (misconfigured upstream ref), not a fast-forward conflict ? I'm proceeding read-only on the local checkout, which is fine for synthesis. I have what I need on Gemini's profile. Let me read the specific section 3 lines that were omitted to ground the precision/FP claims precisely.

8. user (2026-06-19T16:34:07.412Z)

33 | clip (golden) | system | found | tIoU@0.5 P/R/F1 | tolF1 | hits | false+ | missed |
| \?start\ | \?end\ |
34 | ---|---|---|---|---|---|---|---|---|
35 | **GX010141** (29, hard/drilling) | LOCAL | 22 | 0.59/0.45/**0.51** | 0.43 | 13 | 11 |
16 | 0.78 | 0.68 |
36 | | GEMINI | 16 | 0.94/0.52/**0.67** | 0.53 | 15 | 4 | 14 | 0.31 | 0.46 |
37 | **GX010137** (34, normal match) | LOCAL | 39 | 0.77/**0.88**/**0.82** | 0.68 | 30 | 14
| 9 | 0.39 | 0.44 |
38 | | GEMINI | 27 | 1.00/0.79/**0.89** | 0.89 | 27 | 0 | 7 | 0.50 | 0.52 |
39
40 - **Corpus aggregate (n=13, milestone):** tIoU F1@0.5 **0.474** [0.359, 0.579], F1@0.25
0.633; R2 tolF1 0.359. GX010137 (0.822) is the corpus's **best** clip; GX010141 (0.510) is above
the mean. `[verified]`
41 - **Opposite failure modes ? the central finding (n=2).** **LOCAL = high recall, low
precision; GEMINI = high precision, low recall.** On the normal clip GX010137, local's **recall
0.88 BEATS Gemini's 0.79** (local finds *more* real rallies) but local fires 14 false positives;
Gemini fires **zero** false positives but misses 7 rallies. They are **complementary** (on
GX010141 they overlap on only ~3 rallies yet each hits ~13?15 of the golden set). ? **fusion
(local's recall + Gemini-style precision filtering) is the clear win.**
42 - **The gaps, re-prioritized by the data:**
43 - **G2 ? precision is LOCAL's #1 gap** (over-fires on both clips: 11/22 and 14/39
false positives). This is the cheapest + highest-leverage fix (Gate-A / colocation). Filter
local's FPs on GX010137 and it lands ~0.88+ ? at-par with Gemini, since local already out-
recalls it.
44 - **G1 ? recall** is clip-dependent: a local *strength* on normal play (? Gemini), but
on hard continuous-drilling clips *both* miss ~half (GX010141 recall ~45?52%). The recall lever
matters mainly for the hard clips.
45 - **G3 ? rally-END over-extension** (quick-return-after-point): shuttle keeps moving ?
window runs long / merges; R4 (shuttle-only) can't catch it. Contributes to G2's false
positives/merges.
46 - **Local's gap to Gemini SHRINKS on normal play:** hard clip ??0.16 (0.51 vs 0.67) ?
normal clip ??0.07 (0.82 vs 0.89), and local *out-recalls* Gemini there. The deficit is
precision, not detection.
47 - **What is NOT the gap:** boundary precision. When local hits a rally its edges are
tight (|?start|/?end| ? ~0.8 s, on GX010137 even tighter than Gemini). The R2/R3/R4 windowing
is doing its job. `[verified]`
48 - **Long-rally lens refinement (P7, n=15, `[verified]`):** filtering the eval to >5s
rallies (the eye-catching highlights) shows local's single-court over-fire is **short-FP-
dominated** (seg_ratio 1.43?1.04 ? duration-filterable), but **multicourt still over-fires 1.58x
on LONG rallies** (adjacent-court games are genuine long rallies). ? the multicourt precision
gap needs **court-localization (Gate-B / court mask, Lever C)**, NOT a duration filter; and the

strict tolerance precision is gated by **over-merge contamination** of the long-pred set, not just short FPs. See QUALITY_ITERATIONS.md "P7".

9. assistant (2026-06-19T16:34:53.245Z)

I now have everything I need, fully grounded. The Gemini profile (high precision, ~0 false positives, lower recall; G2 = precision is local's #1 gap) is confirmed at lines 41-48, and the multicourt over-fire on long rallies needing court-localization (Gate-B / court mask) is confirmed at line 48. Here is the report.

Prominent-Court Gate ? Multicourt Golden-Clip Quality Synthesis

The **prominent-court gate** (the G2 multicourt precision lever) drops detected rally windows whose shuttle mass falls majority-outside a **recall-safe polygon** containing 100% of golden-rally shuttle medians (`min_in_court_frac=0.5`). Below: per-clip results, the aggregate win, honest null/marginal flags, and a grounded "how close to Gemini" framing.

(a) Per-clip results

clip	n_gt	base P	base R	base F1	gated P	gated R	gated F1	?P	?R	?F1	#dropped	#drop-were-FP	%golden-in-poly
Boxhill_Doubles	43	0.322	0.651	0.431	0.418	0.651	0.509	0.096	0.078	0.20	20 / 20	100%	
mahadevpura_2	40	0.238	0.250	0.244	0.263	0.250	0.256	0.025	0.013	0.04	4 / 4	100%	
GX010128	52	0.494	0.769	0.602	0.494	0.769	0.602	0	0	0	0	?	100%
Badminton_BXH_2	44	0.346	0.614	0.443	0.346	0.614	0.443	0	0	0	0	?	100%

(tolF1: Boxhill 0.262?0.309; mahadevpura 0.122?0.128; GX010128 / BXH_2 unchanged.)

(b) Aggregate quality win

metric	mean across 4 clips	mean across the 2 clips that moved
?P	0.030	0.061
?F1	0.023	0.046
?R	0.000	0.000
dropped windows	24 total	?
dropped that were FP	24 / 24 (100%)	?

The signature is **precision up, recall dead-flat, every dropped window a verified false positive** ? exactly the multicourt G2 behavior the gate is designed for. The headline clip is **Boxhill_Doubles**: P **0.096** / F1 **0.078** by cutting **20** genuine adjacent-court windows (all with median shuttle $x > 0.80$, all temporally distinct from the prominent court's rallies), with the **same 28 golden rallies** matched before and after ? recall flatness verified at the matching level, not merely numerically. The polygon edge ($x = 0.80$) is the sweep-confirmed tightest recall-flat optimum, not curve-fit. This is the gate working as intended: a true spatial separation of a real second court.

(c) Honest null / marginal flags

- **GX010128 ? true NULL, not masked.** Exact no-op (0 dropped). The genuine adjacent-court-looking cluster ($x > 0.87$) is a **stuck-tracker pixel-lock** (284 points at exactly $x = 1680$ px across 291 distinct seconds), not a live court ? it never forms a velocity window, so there is no FP for the gate to remove. Min in-court fraction across all 81 windows is $0.62 > 0.5$, so nothing **can** drop at threshold. No win, no harm.
- **Badminton_BXH_2 ? true NULL.** No-op (0 dropped). The 51 FPs are spatially **co-located** with the true positives inside the prominent court (FP median $x[0.060, 0.497]$? TP $x[0.098, 0.483]$); ~41% of points are a static center-court tracker artifact. No separable second court exists, and

every tighter polygon hits real rallies before FPs (x?0.48 drops 2 TP vs 1 FP). A spatial gate structurally cannot help here.

- ****mahadevpura_2 ? REAL but MARGINAL (+0.013 F1).**** The 4 dropped windows are all confirmed FPs (pre-first / post-last-rally and inter-rally edge fires, zero golden overlap), but the golden medians form ****one central blob with no separable secondary court**** ? the gate is trimming diffuse frame-edge scatter, not performing true multicourt separation. 32/42 baseline FPs are on-prominent-court boundary/over-merge fires the gate ****cannot**** touch. Near-no-op on this clip's dominant FP mode.

****Bottom line on honesty:**** the gate delivers a clean structural win on exactly ****1 of 4**** clips (Boxhill), a marginal-but-real edge trim on 1 (mahadevpura), and ****honest nulls on 2**** (GX010128, BXH_2). ****No clip cost any recall**** (?R=0 everywhere, verified interval-for-interval), and ****no number is inflated**** ? every reported value was reproduced to ?4 decimals against the raw clips. The aggregate ?F1 is modest precisely because the gate only fires where a genuinely separable court exists; it is recall-safe by construction and silent where it has no lever.

(d) How close to Gemini

****There are no Gemini predictions for these four multicourt clips, so a direct number is impossible.**** Inventing one would violate the attribution rule. Instead, reasoning from Gemini's established corpus profile (`\RALLY_DETECTION_GAP_CLOSURE.md` §3, \[verified]` n=2 golden`):

- Gemini's signature is ****high precision (~1.00), near-zero false positives, lower recall**** (e.g. GX010137: Gemini P=1.00 / 0 FP / R=0.79 vs local P=0.77 / 14 FP / R=0.88). ****Precision is local's #1 gap (G2)**** ? local already out-recalls Gemini on normal play, so closing the gap means ****removing false positives, not finding more rallies.**** §3 P7 specifically flags that the residual multicourt over-fire (1.58x on long rallies) needs ****court-localization (Gate-B / court mask)**** ? i.e. ***this exact gate***.

- The prominent-court gate moves local in precisely Gemini's direction ? it strips adjacent-court FPs while holding recall ? so it is directionally the right lever. Distance to a ****perfect-precision reference (P?1.0)**** after gating:

clip	gated P	residual gap to P=1.0	FP-removal progress
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GX010128	0.494	**?0.506**	0 of 41 FPs removed
Badminton_BXH_2	0.346	**?0.654**	0 of 51 FPs removed
Boxhill_Doubles	0.418	**?0.582**	20 of 39 FPs removed
mahadevpura_2	0.263	**?0.737**	4 of 32 FPs removed

Even on its best clip (Boxhill), the gate removes only ****~half**** the false positives and leaves precision at ****0.418 ? a 0.58 residual gap**** to perfect precision. The remaining FPs are ****on-court**** (over-merge, rally-END over-extension G3, on-prominent-court boundary fires), which a ***spatial*** court gate cannot reach. So the gate is a ****partial down-payment**** on the G2 precision gap, not a closer of it.

- ****A real comparison requires running Gemini on these four clips and scoring it against the same golden labels**** ? that is the explicit follow-up. Until then any "X points from Gemini" figure would be fabricated. What can be stated honestly: the gate shifts local ****toward**** Gemini's failure mode (fewer FPs, recall preserved) but, on a perfect-precision yardstick, the bulk of the gap remains and is ****not spatially separable**** ? pointing to fusion / on-court precision rules (Gate-A net-crossings, R4 end-rule) as the complementary next levers.

Verdict

The prominent-court gate is a ****worthwhile but narrow**** multicourt precision lever: it is ****recall-safe by construction**** (100% golden-in-polygon, ?R=0 on every clip) and delivers a ****clean, verified structural win where a genuinely separable adjacent court exists**** (Boxhill: P +0.096, F1 +0.078, 20 real adjacent-court FPs cut). It honestly does ****nothing**** where the second "court" is a tracker artifact (GX010128) or where FPs are co-located on the prominent court (BXH_2), and only ****marginally**** trims edge scatter where the golden mass is one blob (mahadevpura). Because it is free of recall cost and zero-harm on its nulls, it is a ****safe, ship-able down-payment on G2**** ? but it closes only the ***spatially-separable*** slice of the

precision gap (?half the FPs on its one strong clip), so it should land ****flag-gated/default-OFF** per the Launch-Report discipline** and be paired with on-court precision levers (Gate-A, R4 end-rule, fusion) and an actual Gemini run on these clips to quantify the true remaining distance.

Source grounded: `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\RALLY\_DETECTION\_GAP\_CLOSURE.md` S3 (lines 33?48) for Gemini's profile; per-clip rows from the verified JSON provided. Note: `git pull --ff-only` failed on this branch (upstream is misconfigured to a non-existent ref `docs/court-floor-rally-end`), so this synthesis used the local checkout read-only ? no Gemini numbers were invented.